

Field Trials of Restoration-Capable Quantum Key Distribution Networks with Optical Routers

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Abstract We demonstrate physical-layer failure detection and automated recovery in quantum key distribution networks utilising commercial prepare-and-measure BB84 and polarisation entanglement-based BBM92 systems integrated with self-developed all-transparent optical routers over a field-deployed network testbed. ©2026 The Author(s)

Introduction

Quantum key distribution (QKD) [1], with the promise of information-theoretic security, has made significant progress over the past decades, evolving from laboratory demonstrations to commercial-grade deployments. However, point-to-point (P2P) QKD alone is insufficient for practical operations in real communication networks. P2P configurations implicitly assume a static and dedicated optical path between two sites, an assumption that does not hold in operational optical network infrastructures.

In carrier-grade networks, redundancy and protection mechanisms are essential to maintain service continuity against anomalies such as fibre degradations or cuts. Accordingly, QKD networks (QKDNs) must be evaluated not only under static P2P conditions but also in the context of realistic network operations that employ working and protection paths [2]. In such scenarios, the ability to switch optical routes is crucial. Recently, optical switching and routing have been introduced in various laboratory experiments and field trials of QKDNs to enhance network reconfigurability [3–7]. However, to the best of our knowledge, no studies on the failure restoration of QKDNs have been reported so far.

In this paper, we experimentally demonstrate an automated switchover scheme between working and protection path during a fibre degradation event. We characterise key aspects of these redundancy and protection mechanisms, specifically their ability to preserve the quantum states used for information encoding in QKD systems, as well as how quickly the QKD link can recover and resume secure key generation after a fibre route switch, since these factors directly impact the fea-

sibility of integrating QKD into operational optical networks. The experiments are conducted on a field-deployed QKDN testbed using commercial QKD systems from Toshiba, ID Quantique, and S-Fifteen Instruments, providing practical insights into the development of resilient quantum networks in real-world environments.

Field Trial Setups

In this work, we carry out three field trials using the Toshiba, ID Quantique, and S-Fifteen Instruments QKD systems over the National Quantum-Safe Network (NQSN) [8] testbed in Singapore. A pair of self-developed optical routers, previously tested for compatibility with the quantum channels, are synchronised to switch between the working and protection paths at Alice and Bob, for both the classical (C) and quantum (Q) channels, or for the quantum channel alone (Fig. 1). An electrically-driven variable optical attenuator (VOA) is used to emulate path failure by gradually increasing attenuation in the Q channel. Once the preset failure threshold is reached, the path switchover is triggered.

The switching decision, made by the optical routers deployed on the same network as the QKD management network, is based solely on link quality metrics accessible via the APIs for the QKD systems. This enables the switchover scheme to be implemented without any modifications to the existing QKD hardware or software.

The first field trial (Fig. 1(a)) involved the Toshiba Long Distance QKD system based on prepare-and-measure (PM) phase-encoding decoy states BB84. Both Alice and Bob are located in the NQSN central node at the Centre for Quantum Technologies (CQT), National University of Singapore (NUS). There is a working path and a pro-

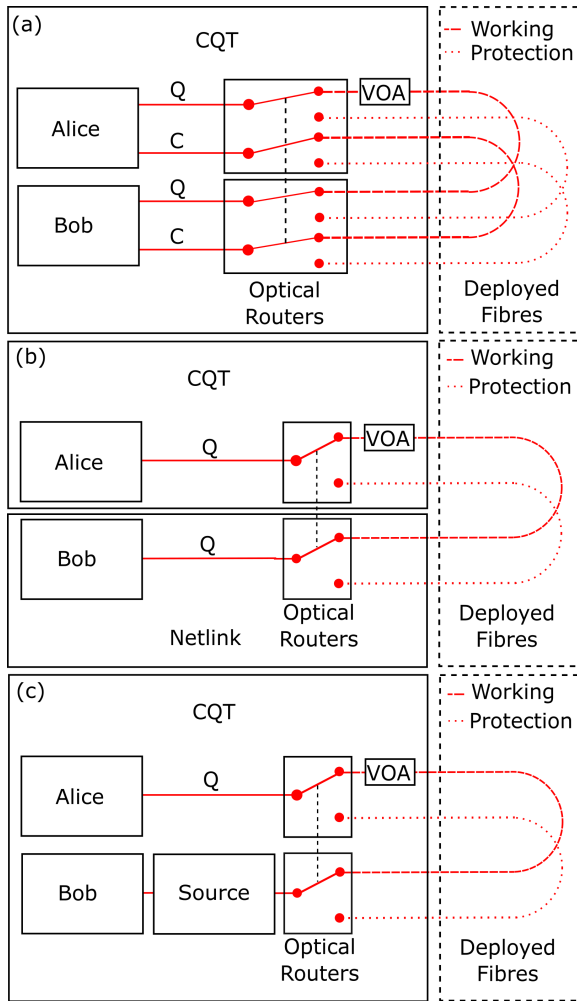


Fig. 1: Topology of the automated switchover field trials for the QKD Systems. (a) Toshiba. (b) ID Quantique. (c) S-Fifteen.

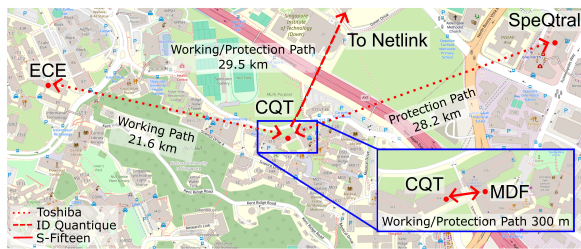


Fig. 2: Fibre paths of the field tests of the three QKD systems. Map data © OpenStreetMap contributors.

tection path for both quantum and classical channels (Fig. 1(a)). In each path, Q and C channels are separate fibres in the same fibre-optic cable, thus having the same fibre length. The working path is a 21.6 km loop-back fibre link via NUS Department of Electrical and Computer Engineering (ECE), whilst the protection path is a 28.2 km loop-back fibre link via SpeQtral Pte. Ltd. outside the NUS campus. In nominal operation, the raw keys are being accumulated to fill a key block. Once a key block is filled, the raw keys are sent for post-processing to generate the final secret keys. We define a failure threshold as the average key block fill rate over the previous 2 minutes being $< 0.0003 \text{ s}^{-1}$. The dual switchover of Q and

C channels is necessary because Toshiba QKD systems require both channels to have the same fibre length to ensure correct synchronisations and post-processing.

The second field trial (Fig. 1(b)) involved the ID Quantique Clavis XGR QKD System also based on PM phase-encoding decoy states BB84. Alice is located at the CQT node, while Bob at a Netlink Trust node outside the NUS campus in the north of Singapore. Due to the limited availability of fibre routes, only the working and protection paths for the Q channel are deployed. (Fig. 1(b)). Both Q and C channels are carried in different fibres along the same fibre route with a distance of 29.5 km. The timing synchronisation still works despite the extra length of $\sim 8 \text{ m}$ on the Q channel introduced by the optical routers. Under nominal operation, the average total count rate detected by Bob, measured over a 5-minute interval, is $57,742 \text{ s}^{-1}$. The failure threshold is defined as 50% of this value, i.e., $28,871 \text{ s}^{-1}$.

The third field trial (Fig. 1(c)) involved the S-Fifteen polarisation entanglement-based BBM92 QKD system [9, 10]. The working and protection paths are deployed only for the Q channel (Fig. 1(c)), between the entangled photon-pair source (EPPS) and Alice, as timing information for synchronisation is derived from the detection times at Alice and Bob, a separate optical link is not required. Bob is connected to the EPPS via a back-to-back fibre connection. The EPPS, Alice, and Bob are all located in the CQT node. The working and protection paths between the EPPS and Alice are loop-back fibre links with a length of 300 meters via a main distribution frame (MDF) room near the edge of the CQT building (Fig. 2). The total singles detector count rate of Alice is monitored and a failure threshold of $< 50 \text{ k/s}^{-1}$ is used for the trial.

Field Trial Results

In the first field trial with the Toshiba system (Fig. 3), as the attenuation in the VOA rises, the key block fill rate drops. A switchover is triggered when key block fill rate falls below the failure threshold of 0.0003 s^{-1} (Fig. 3(b)). The QKD link is recovered 47 minutes after the switchover (Fig. 3(c) and (d)). The required re-synchronisation after the switchover significantly increased the recovery time.

In the second field trial with the ID Quantique system, with the increase in the VOA attenuation, the total detected count rate falls below the failure threshold of $28,871 \text{ s}^{-1}$ and a switchover is triggered (Fig. 4(a)). It can be observed that the telemetry on the total detected count rate is still available during the re-synchronisation after the switchover (Fig. 4(a)). It took 19 minutes for the QKD link to be fully recovered after the switchover (Fig. 4(b)),

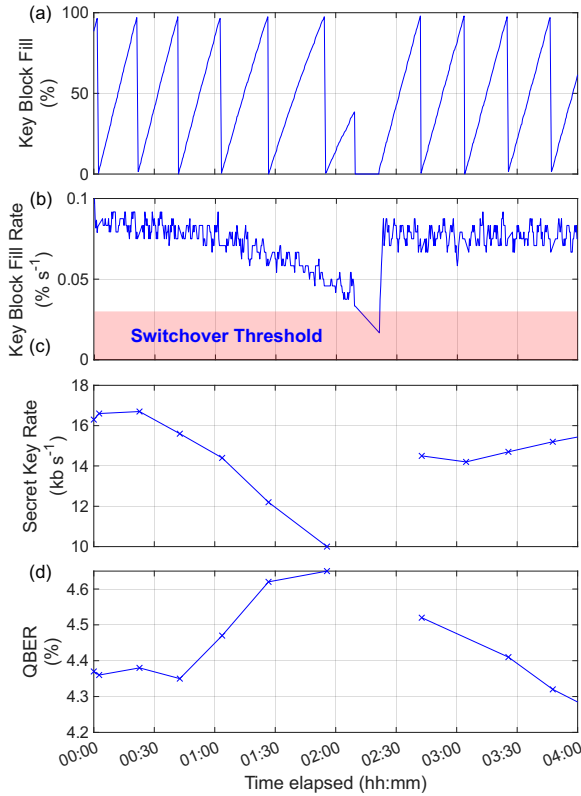


Fig. 3: Metrics during switchover for Toshiba system

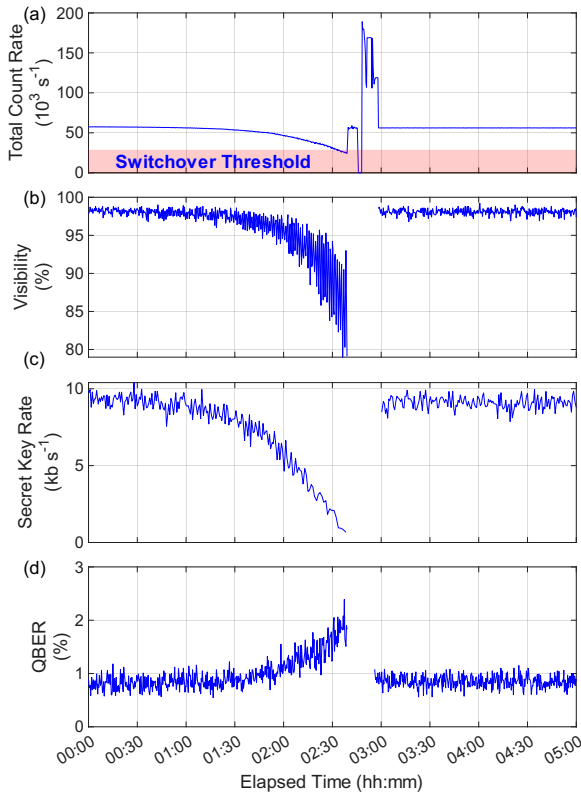


Fig. 4: Metrics during switchover for ID Quantique system

(c) and (d)).

In the third field trial with the S-Fifteen system, it is clearly observed from Fig. 5(a) that singles count rate in Alice steadily decreases due to the increase of the attenuation in the Q channel. Once singles count rate of Alice falls below the thresh-

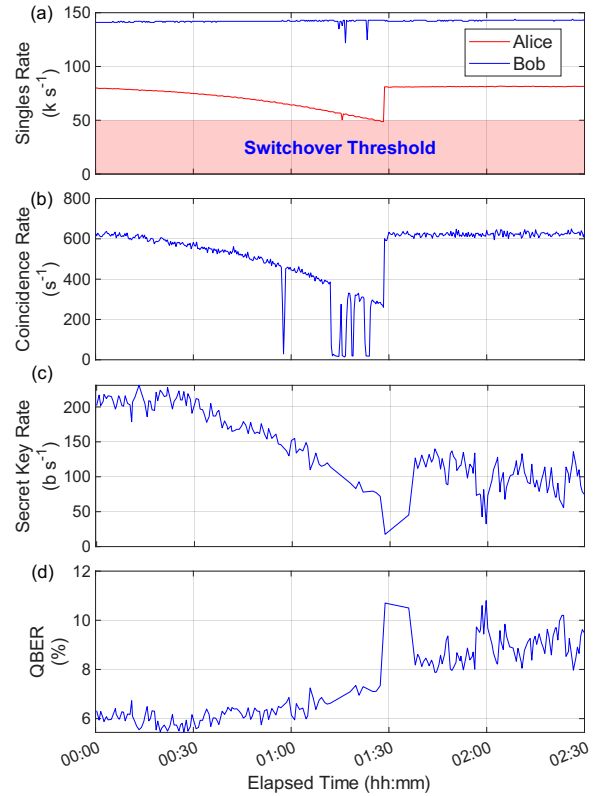


Fig. 5: Metrics during switchover for S-Fifteen system

old of $< 50 \text{ ks}^{-1}$, the switchover is triggered. The singles count rate of Alice increased slightly after the switchover compared to the start of the test, due to differences in loss between the working and protection paths. As shown in Fig. 5(b), coincidence rate between Alice and Bob has a similar trend to that of singles count rate of Alice. It took 9 minutes for the QKD link to be fully recovered after the switchover (Fig. 5(c) and (d)).

In all three field trials, the introduction of the optical routers in the quantum channel lead to a negligible increase in the QBER (Fig. 3(d), 4(d), and 5(d)), demonstrating transparency to the quantum states, encoded in phase and polarisation, used in QKD systems.

Conclusions

We experimentally investigate the physical-layer resilience in QKDNs using commercial BB84 and BBM92 QKD systems integrated with optical routers in a field-deployed testbed. Automatic failure detection and fibre path switching are achieved using QKD-specific performance metrics without modifying QKD hardware or software. Measured recovery times are 47 minutes and 19 minutes for the BB84 systems and 9 minutes for the BBM92 system, highlighting the impact of QKD system synchronisation mechanisms on the restoration latency. These results provide practical guidance for designing scalable and resilient quantum-secure optical networks.

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